Stage VI-0: Hierarchical Physical World-Model Validation

Strictly offline local smoke validation

2026-07-12

1 Decision

Decision C - descriptor schema still incomplete is the single main decision. The hierarchy resolves invalid-field mixing, but the full schema and uncertainty gates are not yet validated.

2 Scope

Frozen V3/V4 CSVs only. No exact solver, Delta-q search, active learning, HPC, or data edit. No learned state is transferable.

3 Hierarchy

Level 1 models SC existence and thermodynamics. Level 2 models the SC branch. Level 3 models trusted spectra. Level 4 models trusted gapped-SC Pfaffian outputs. Phase/topology labels are evaluation-only.

4 Matched smoke results

The mean spatial SC AUROC is 0.979. Median spatial R^2 is 0.441 for Level 1, 0.797 for Level 2, 0.051 for Level 3, and 0.912 for Level 4. Thermodynamics and $|q|$ are learnable. P_0/P_π are learnable after trusted-gapped conditioning, but log-gap, k_{gap}, Pfaffian margin, and uncertainty-error alignment remain weak. Hierarchical Level-3 NLL improves over the same-split unified model; Level-4 improvement is not universal. Probe uncertainty is not monotonic across all run/level trajectories.

5 Critical answers

Thermodynamics can be learned independently. Conditioning makes q learnable, but ΔF_{UF} cannot be tested. Topology regression becomes meaningful after invalid points are removed, yet its uncertainty is not consistently calibrated. The remaining failures are not mask sparsity alone because the spatial spectral/topology tests contain 639–694 valid points.

6 Caveat

The common frozen schema has no reliable ΔF_{UF} field.

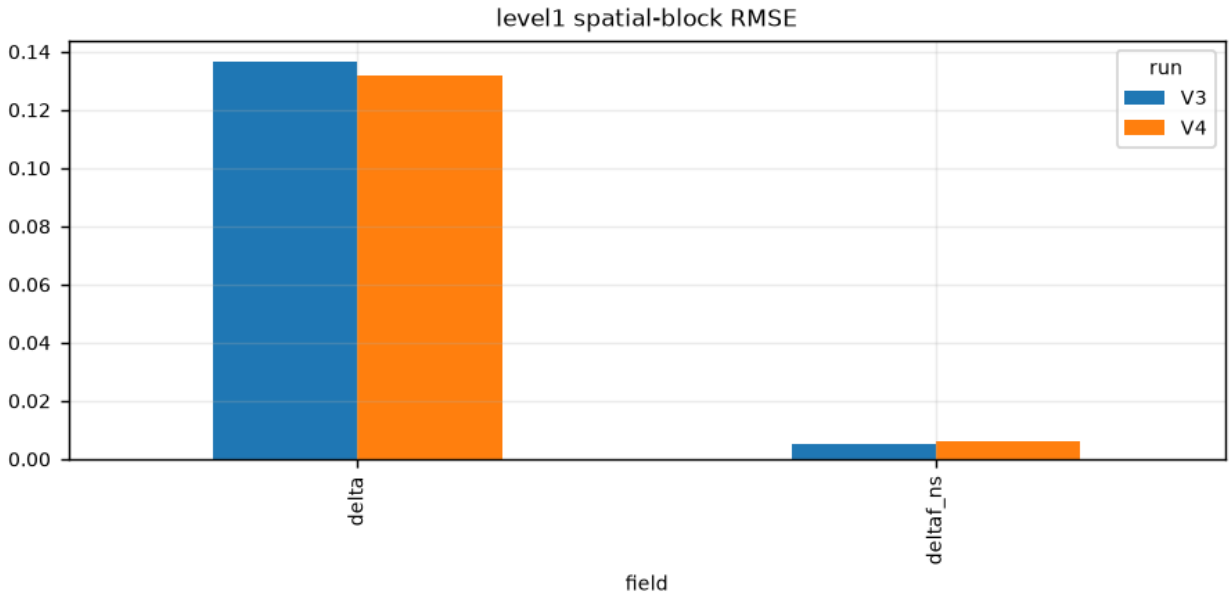


Figure 1: figure 01 level1 thermodynamics

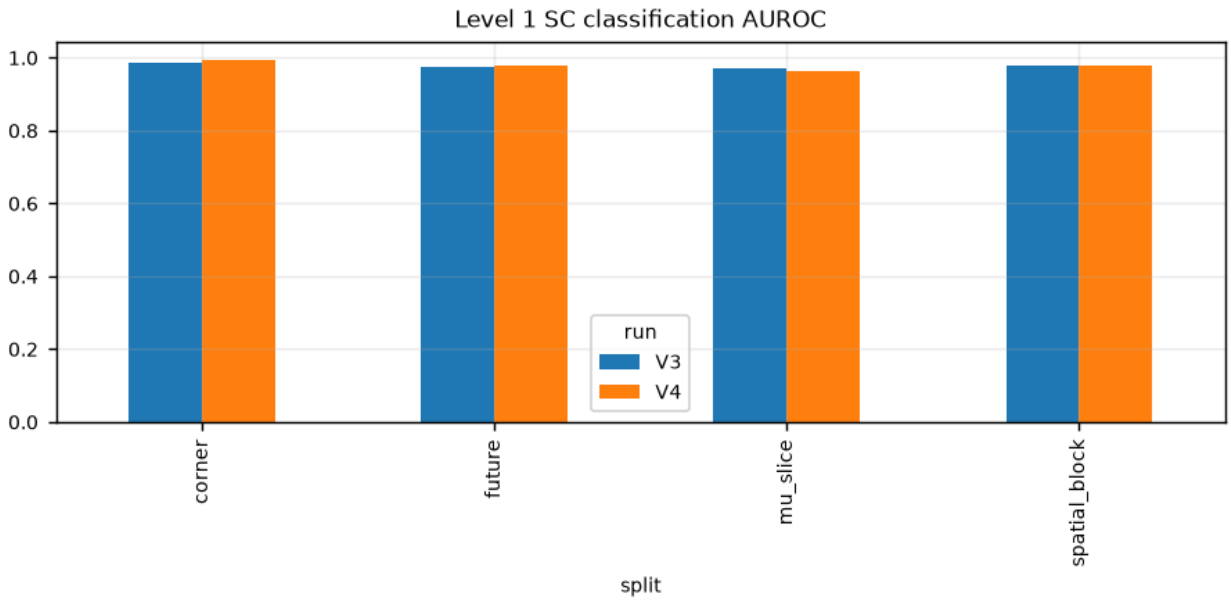


Figure 2: figure 02 level1 classification

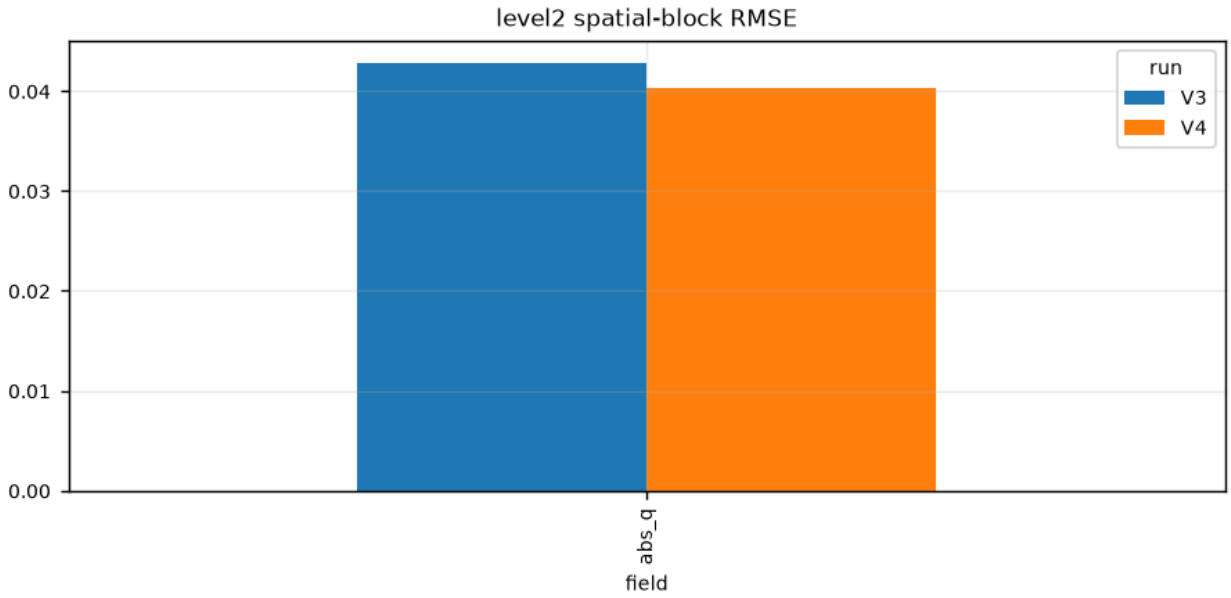


Figure 3: figure 03 level2 branch

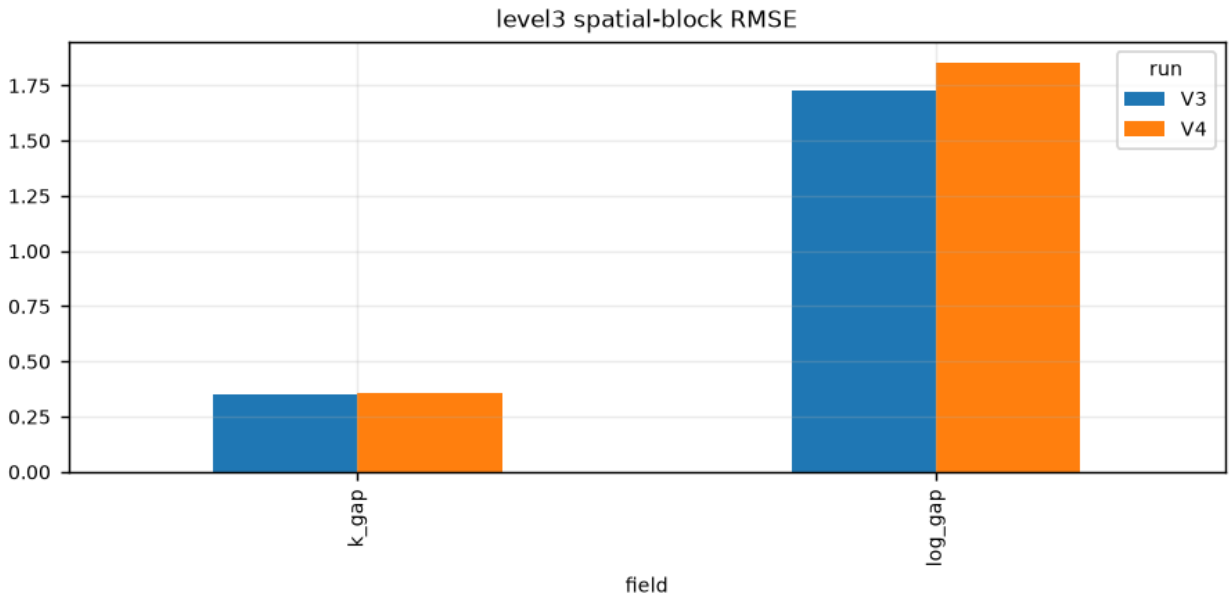


Figure 4: figure 04 level3 spectral

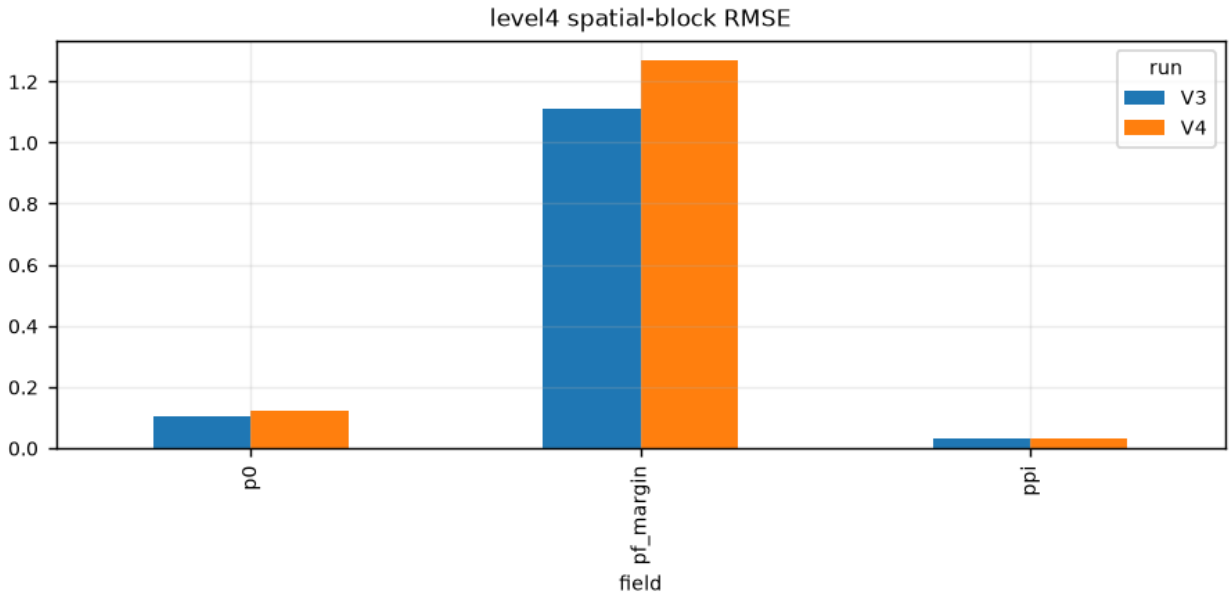


Figure 5: figure 05 level4 topology

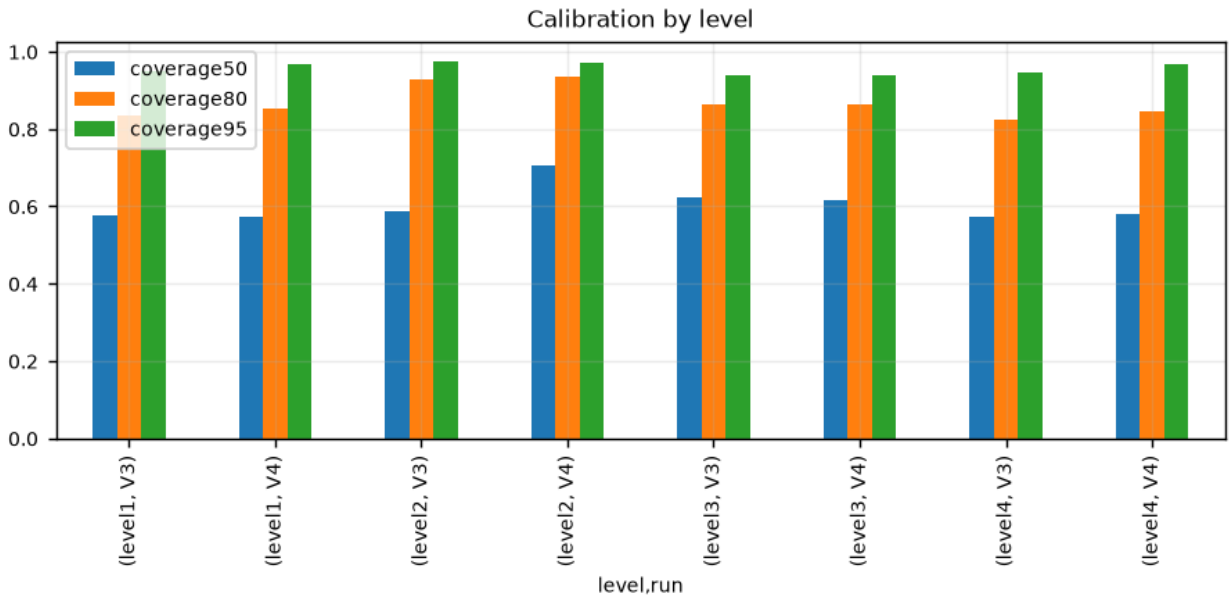


Figure 6: figure 06 calibration by level

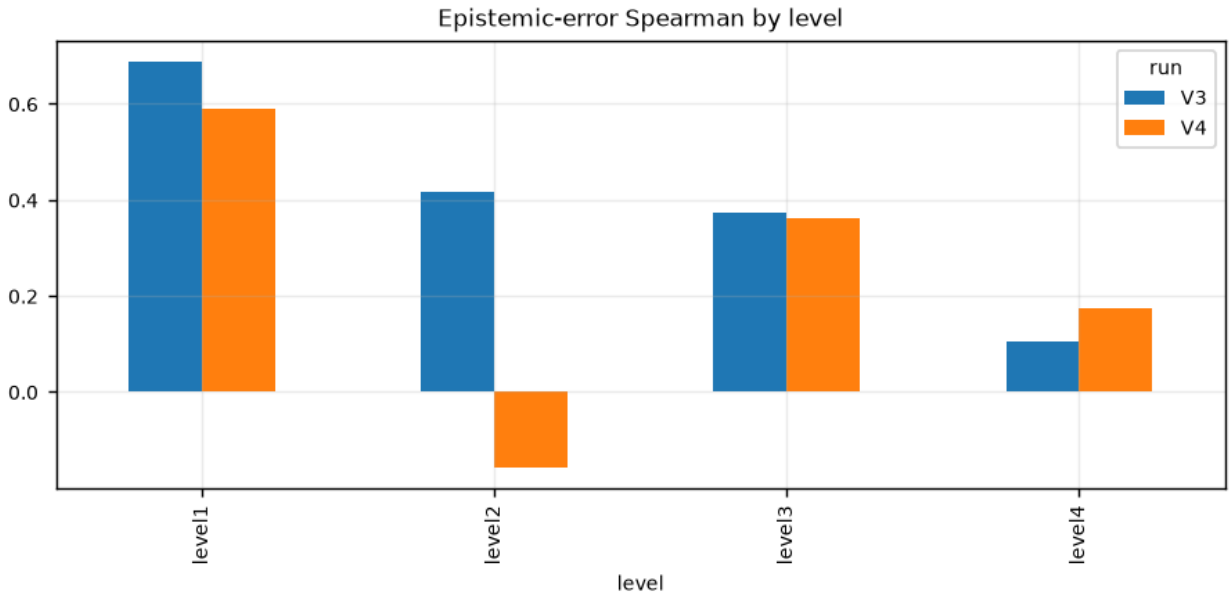


Figure 7: figure 07 uncertainty error by level

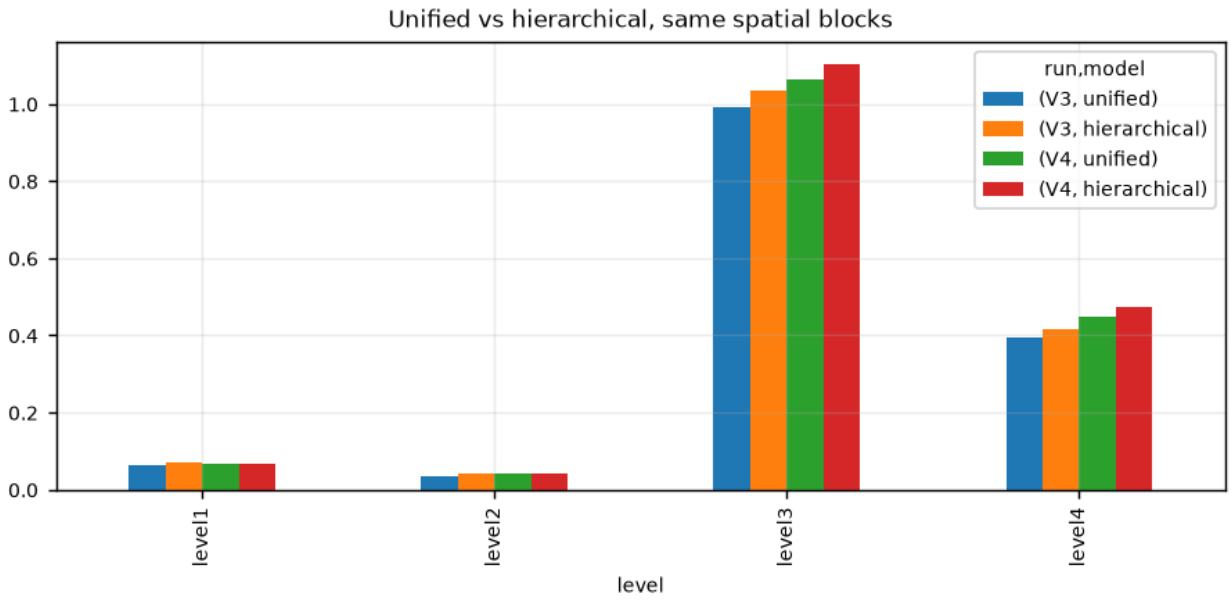


Figure 8: figure 08 unified hierarchical

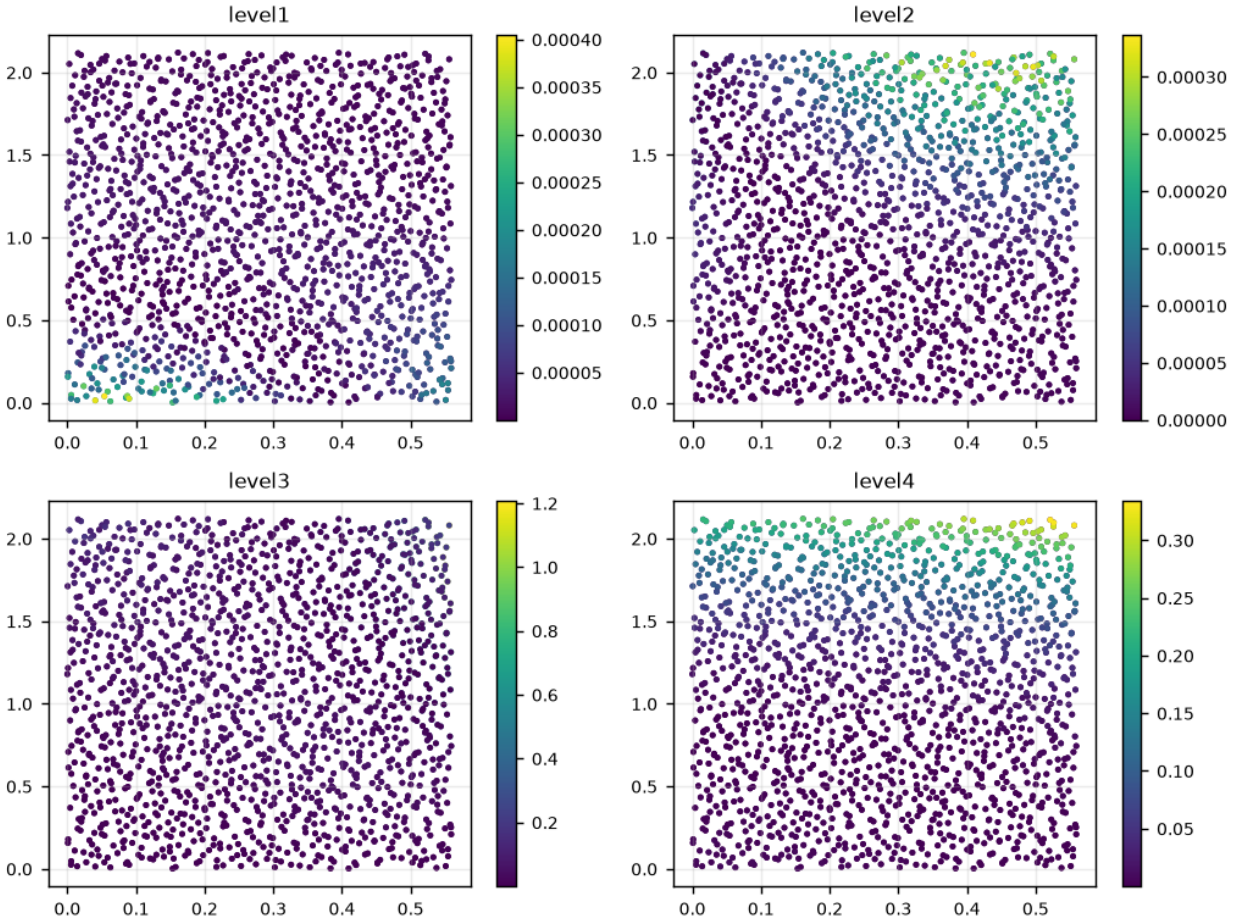


Figure 9: figure 09 probe maps

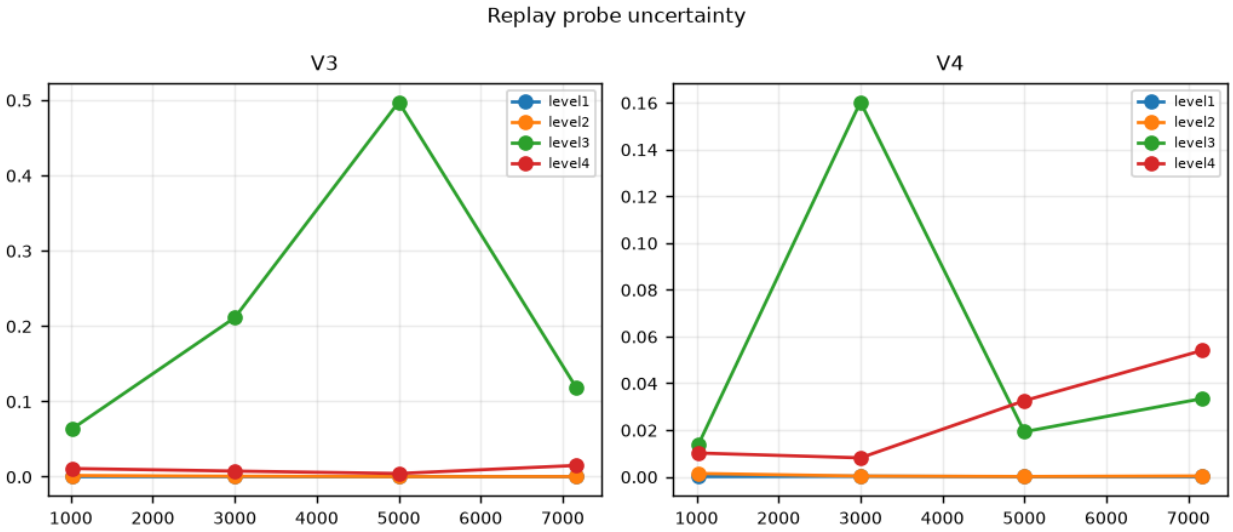


Figure 10: figure 10 replay uncertainty

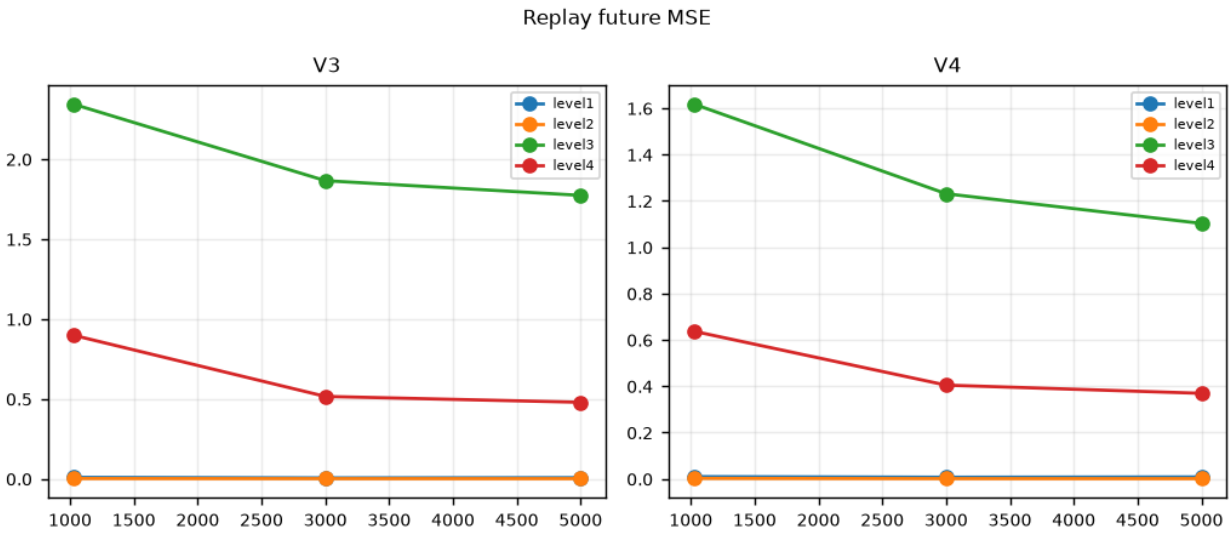


Figure 11: figure 11 replay future error

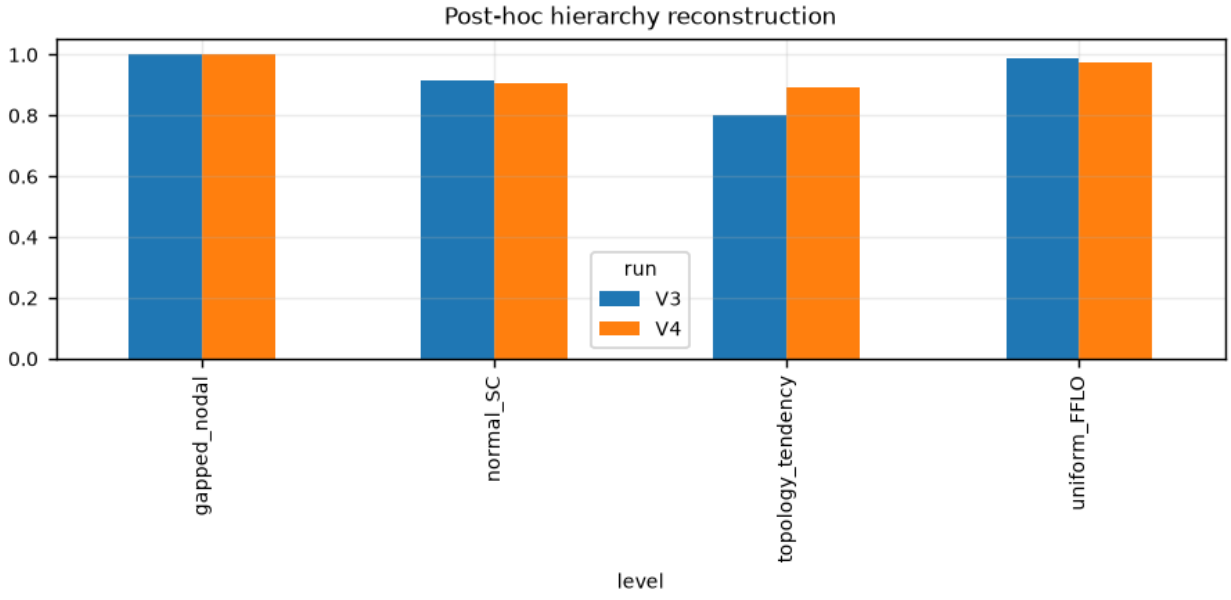


Figure 12: figure 12 phase reconstruction

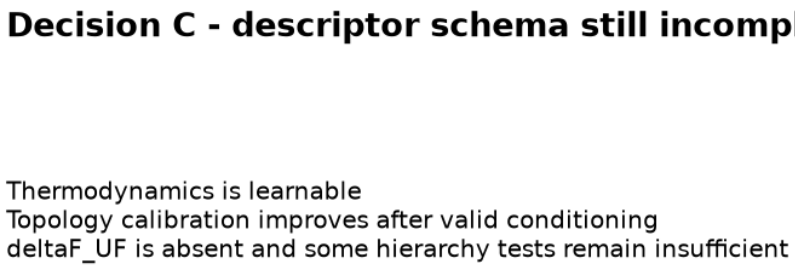


Figure 13: figure 13 decision

7 Isolation

Only schema, masks, fixed transforms, architecture/loss source, validation source, and tests may be reused. All learned state is denied.